

# Modeling Loci and Intersections with Conic Sections

## Algebra 2 — Conics and Modeling

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Circles built from a distance condition, parabolas built from a focus and a directrix, intersections of a line with a conic, and intersections of two conics. Two ideas run through the whole sheet:

- **A locus is a sentence about distance.** Translate the sentence into a distance equation, square both sides, and simplify. A circle is “distance to one point is constant”; a parabola is “distance to a point equals distance to a line”.
- **An intersection is a system.** Substitute one equation into the other, solve the single-variable equation that results, then substitute back for the partner coordinate. Report ordered pairs, not just  $x$  values.
- Reference forms:  $(x - h)^2 + (y - k)^2 = r^2$  for a circle with center  $(h, k)$ ;  $x^2 = 4py$  for a parabola with vertex at the origin, focus  $(0, p)$  and directrix  $y = -p$ .

1. A circle has center  $(3, -2)$  and radius 5.

Evaluate  $(x - 3)^2 + (y + 2)^2$  at the point  $(7, 1)$ , and state whether that point lies on the circle.

Answer: \_\_\_\_\_

2. A circle is the set of all points 5 units from the origin, so its equation is  $x^2 + y^2 = 25$ . Find both  $y$  values on this circle when  $x = 3$ .

Answer: \_\_\_\_\_

**3.** A parabola has focus  $(0, 3)$  and directrix  $y = -3$ , so  $p = 3$  and its equation is  $x^2 = 4py$ . Find the  $y$  coordinate of the point on this parabola whose  $x$  coordinate is 6.

Answer: \_\_\_\_\_

**4.** For the same parabola from problem 3,  $x^2 = 12y$ . Find both  $x$  values on the parabola when  $y = 12$ .

Answer: \_\_\_\_\_

**5.** Find every point where the line  $y = x + 1$  meets the circle  $x^2 + y^2 = 25$ . Give your answers as ordered pairs.

Answer: \_\_\_\_\_

6. Complete the square to rewrite

$$x^2 + y^2 - 6x + 8y - 11 = 0$$

in center-radius form, then state the center and the radius.

Answer: \_\_\_\_\_

7. Find every point where the circle  $x^2 + y^2 = 25$  meets the parabola  $y = x^2 - 5$ . Give your answers as ordered pairs.

Answer: \_\_\_\_\_

**8.** Find every point where the line  $y = 2x - 3$  meets the parabola  $y = x^2 - 2x + 1$ , and say what the number of solutions tells you about how the line sits against the curve.

Answer: \_\_\_\_\_

**9.** A point  $P(x, y)$  moves so that it is always the same distance from the point  $(0, 5)$  as it is from the line  $y = -5$ .

(a) Starting from  $\sqrt{x^2 + (y - 5)^2} = y + 5$ , derive the equation of the locus and simplify it.

(b) Using your equation, find both points on the locus with  $y = 5$ .

(b) \_\_\_\_\_

**10.** Find every point where the circle  $x^2 + y^2 = 25$  meets the hyperbola  $xy = 12$ . Give your answers as ordered pairs.

Answer: \_\_\_\_\_

**11.** A point  $P(x, y)$  moves so that its distance from  $A(6, 0)$  is always twice its distance from  $B(0, 0)$ .

Write  $PA = 2 \cdot PB$  as a distance equation, square both sides, and simplify until the locus is in center-radius form. State the center and the radius.

Answer: \_\_\_\_\_

**12. Synthesis challenge.** The cross-section of a satellite dish is the parabola  $x^2 = 8y$ , whose focus is  $(0, 2)$  and whose directrix is  $y = -2$ . A circular guard ring around it has equation  $x^2 + y^2 = 20$ .

(a) Find every point where the ring meets the dish.

(a) \_\_\_\_\_

(b) Take the intersection point with positive  $x$ . Find its distance from the focus  $(0, 2)$ , and separately find its distance from the directrix  $y = -2$ .

(b) \_\_\_\_\_

(c) Explain what your two answers in part (b) say about every point of the dish.